

TrES-3b

R-0940

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_260624 · created 2026-07-29T12:53:41+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461215.8089 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.055
Transit depth (Rp/Rs)^2	3.7 +/- 2.13 [%]
Orbital Inclination (inc)	84.43 +/- 2.8
Airmass coefficient 1 (a1)	0.279 +/- 0.024
Airmass coefficient 2 (a2)	1.25 +/- 0.72
Scatter in the residuals of the lightcurve fit is	8.48 %
Best Comparison Star	#2 - [474, 406]
Optimal Aperture	2.11
Optimal Annulus	12.63
Transit Duration (day)	0.0362 +/- 0.0098

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.055 vs 0.16309 (deviation 0.5σ)
Duration fit vs published	0.0362 d vs 0.0567 d (deviation 1.98σ)
Mid-time O–C	3.7 min
Depth SNR	3.3
Residual scatter	8.48 %

Light curve

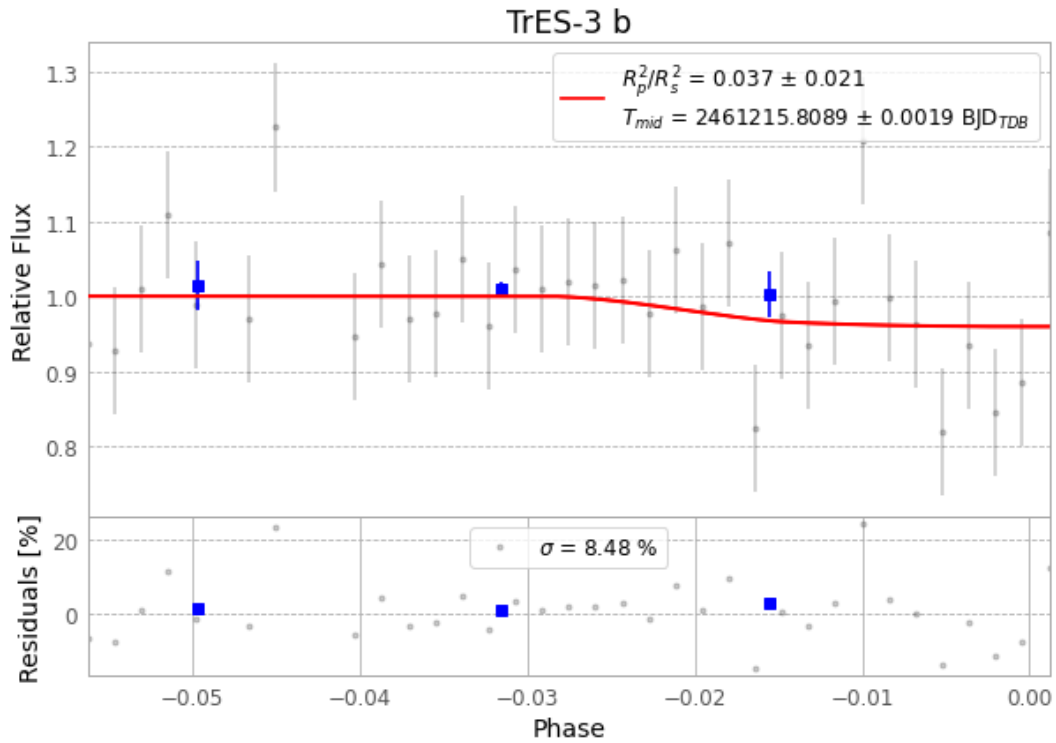


Figure 1 — FinalLightCurve_TrES-3 b_2026-06-23.png (SHA-256 d8ad95538dd8e071...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [230, 191], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0f185b2e1d2bb3ae...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

37 FITS frames (24 MB), TRES-3260624053317.FITS → TRES-3260624072115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-260624.log (a483d64d6596...), FinalLightCurve_TrES-3 b_2026-06-23.png (d8ad95538dd8...), FinalLightCurve_TrES-3 b_2026-06-23.csv (d0466d2b3a9c...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 12:53Z.